

Mst. Fahmida Sultana Naznin

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RESEARCH INTERESTS

My research domain is AI in Healthcare, with a strong focus on Natural Language Processing, Computer Vision, and Bioinformatics. I am particularly interested in integrating demographic and historical data to improve contextual insights.

WORK EXPERIENCE

- **PinkLifeLine**  May 2024 - Present
Founder & Chief Executive Officer Dhaka, Bangladesh
- **Biomedical Science and Engineering Research Center (BIOSE)**  Jul 2024 - Present
Student Researcher Dhaka, Bangladesh
Supervisors: Md Rakibul Hasan, Md. Mehedi Hasan Shawon Project: Radiology Report Summarization
- **Esolute**  Oct 2022 - Feb 2024
Senior Human Resources Coordinator Dhaka, Bangladesh

EDUCATION

- **Bangladesh University of Engineering and Technology** 2022 - Present
B.Sc, Computer Science & Engineering Dhaka, Bangladesh
 - GPA: 3.78/4.00 (Current)

PUBLICATION

C=CONFERENCE, J=JOURNAL

REFEREED CONFERENCE AND JOURNAL PUBLICATIONS

- [C.1] CSTR: Context-Driven Sequential Transfer Learning for Abstractive Radiology Report Summarization, by **Mst. Fahmida Sultana Naznin**, Adnan Ibney Faruq, Mostafa Rifat Tazwar, Md Jobayer, Md. Mehedi Hasan Shawon, and Md Rakibul Hasan. 2025. In Findings of The 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025). DOI:10.18653/v1/2025.findings-acl.1360.

UNDER REVIEW

- [C.1] CBRS: Cognitive Blood Request System with Bilingual Dataset and Dual-Layer Filtering for Multi-Platform Social Streams, by **Mst. Fahmida Sultana Naznin**, Anik Saha, Zia Ul Hassan Abdullah, Anisa Binte Asad, K G Subarno Bithi, and A. B. M. Alim Al Islam. 2025. Manuscript submitted to **ACL ARR, May 2025**.
- [C.2] PAN-ICA YOLO: Pyramid-Aware Neighbor-Only Inter-Channel Attention with Dense Optical Flow Mid-Fusion for Video Object Localization, by Adnan Ibney Faruq, and **Mst. Fahmida Sultana Naznin**. 2025. Manuscript submitted to **WACV 2026**.
- [J.1] A Diffusion-based Feature Map Fusion Model for Spatially Adaptive Retinal Vessel Segmentation, by SM Nuruzzaman Nobel, **Mst. Fahmida Sultana Naznin**, Asmaa Soliman Al-Moisheer, and Mohammad Ali Moni. 2025. Manuscript submitted to **IEEE Transactions on Neural Networks and Learning Systems**.
- [J.2] MedSegNet-X: Hybrid Noise-Enhanced Diffusion based Medical Image Segmentation with ConvNeXt Transformer, by SM Nuruzzaman Nobel, **Mst. Fahmida Sultana Naznin**, Md All Moon Tasir, Asmaa Soliman Al-Moisheer, and Mohammad Ali Moni. 2025. Manuscript submitted to **IEEE Transactions on Multimedia**.

WORKSHOPS & POSTERS

- [1] DDS-E-Sim: A Transformer-based Probabilistic Generative Framework for Simulating Error-Prone DNA Sequences for DNA Data Storage, by **Mst. Fahmida Sultana Naznin**, Swarup Sidhartho Mondol, Adnan Ibney Faruq, Debashmita Saha, Ahmed Mahir Sultan Rumi, and A. B. M. Alim Al Islam. 2025. NeurIPS 2025 Workshop SPIGM.

ONGOING WORKS

Brain Computer Interface

Advisor: **Md. Golam Rabiul Alam**, Professor, Dept. of Computer Science & Engineering, BRAC University

- Foundation Model for Robust Epilepsy Detection with Focus on Asian Demographic Data

May 2025 - Present
[Google Scholar](#)

Advisor: **Dr. A.B.M. Alim Al Islam**, Professor, Dept. of Computer Science & Engineering, BUET

- Contrastive Self-Supervised Learning with Explainable AI for Epilepsy Detection from EEG Spectrograms

Jan 2025 - Present
[Google Scholar](#)

Longitudinal Mammogram Risk Prediction

Advisor: **Dr. A.B.M. Alim Al Islam**, Professor, Dept. of Computer Science & Engineering, BUET

- The More, The Better? Impact of Longitudinal Spatiotemporal Context in Mammograms for Improved Breast Cancer Risk Prediction and Tumor Localization

May 2025 - Present
[Google Scholar](#)

Multimodal Radiology Report Summarization

Advisor: **Md Rakibul Hasan**, PhD Candidate (Computing), Curtin University; Senior Lecturer (on leave), BRAC University

- The More, The Better? Study on Additional Context in Radiology Report Summarisation

Dec 2024 - Present
[Google Scholar](#)

AI-Based Self-Testing Device for Breast Cancer Patients

Advisor: **Dr. M Sohel Rahman**, Professor, Dept. of Computer Science & Engineering, BUET

- Breast Cancer Lump Detection Using Deep Learning on LED Image Data

Oct 2024 - Present
[Google Scholar](#)

PROJECTS

- **PinkLifecycle: [All in one breast cancer care platform]**

Tools: [Spring Boot, React, WebSocket, RabbitMQ, JUnit, TestContainers, Docker, Kubernetes, Terraform, GCP, GitHub Actions]

- Developed AI for mammogram analysis, designed software architecture, and strategized the business model.

May 2024 - Present



- **GreenBlox: [Blockchain based solution for Carbon Credit Market]**

Tools: [Celo, IoT, Smart contracts, React, Node.js]

- Developed ML model with XGBoost for solar irradiance prediction, designed algorithms for carbon reduction in solar plants, and led system architecture design.

Sep 2024



SKILLS

- **Programming Languages:** Python, C, C++, Java
- **Web and Database:** HTML, CSS, React, Oracle, mongoDB
- **Data Science & Machine Learning:** Pandas, Numpy, Matplotlib, Seaborn, scikit-learn, PyTorch
- **Specialized Area:** Natural Language Processing, Computer Vision, Bioinformatics
- **Soft Skills:** Presentation, Problem Solving, Creative Writing, Mentoring & Collaborating

HONOURS AND AWARDS

CS competitions:

- | | |
|------|---|
| 2025 | Blockchain Olympiad Bangladesh (Blockchain & AI Category) - Honorable Mention
2nd ISETS-ESCAP & 3rd ISETS Global Competition of Youth Voice on Energy Transition (Global) - 2nd prize (Fully funded trip to UN-ESCAP Headquarters in Bangkok out of 105 projects)
PITCH PROPEL PROSPER – A Startup Showcase by SustainLaunch Labs - Champion (awarded USD 1000)
Idea Innovation 5.0 by Interactive Cares - 1st Runner Up (awarded USD 400)
Huawei Seeds For The Future Bangladesh - Finalist (Top 21)
2025 Johns Hopkins Healthcare Design Competition (Global) - Champion (awarded USD 5000 out of 440 projects from 203 universities across 44 countries, sponsored by J & J MedTech)
Youth Startup Summit 2025 - 2nd Runner Up (awarded USD 1000 out of 75,000 participants)
Hult Prize On Campus - 1st Runner Up |
| 2024 | ISETS-ESCAP Youth Voice Competition (Global) - South Asia Divisional 2nd (Invited to Youth Dialogue at UN ESCAP, Bangkok — selected from 300 applicants across 77 teams)
IEEE WIE Big Idea Pitch Competition (Global) - Best Impact Prize (awarded USD 500)
Orange Corner Bangladesh presents IDEATION CHALLENGE 3.0 - Champion (awarded BDT 100,000 out of 236 applications from 700+ participants)
Luminous Teen Presents THRIVE INNOVATION CHALLENGE 1.0 - Champion (awarded USD 100)
Lean Sprint 2.0 - 2nd Runner up
Hult Prize On Campus - Finalist (Top 5)
ISCEA Prize Global Case Competition - (awarded USD 770 as 70% scholarship (Top 50))
Blockchain Olympiad Bangladesh (AI Category) - Finalist (Top 16)
Creaventure 3.0 - Finalist (Top 8) |
| 2023 | IEEE WIE Big Idea Pitch Competition (Global) - Best Impact Prize (awarded USD 300)
SympSIST: Humanitarian Project Design Idea Competition in ICCIT - Champion |

IEEE BUP Branch Fest - 2nd Runner up
 Ennovators 7.0 - Finalist (Top-5) (offered an internship out of 35,000 contestants from 153 universities)
 2022 Entrepreneurship Idea Contest & Boot-Camp, INDCON, ICCIT - 2nd Runner up

Research and Fellowships:

2025 BUET UIHP Cohort-5 by Research and Innovation Centre for Science and Engineering
 (May 2025 - August 2025) - awarded USD 300
 2024 Global Changemaker Fellowship Cohort-4 (Sep 2024 - Jun 2025) - top 33 out of 7,000 from 115 countries
 Womentor 5.0 by Banglalink (Jul 2024 - Dec 2024) - top 15 STEM student out of 400+ applicants

Writing Competitions:

2023 National Essay Competition (Digital Bangladesh Fair-2023) - 1st;
 2021 The Queen's Commonwealth Essay Competition (**Global**) - Bronze;
 2020 National Human Rights Essay Competition - 4th (Dhaka Division - 1st);
 2019 Eastern University Essay Competition - 13th;
 2018 Talent Hunt Competition (Dhaka Divisional Best Student in Language and Literature 2018,
 Thana Best Student 2014-2018);
 National Essay Competition on 7th March by Ideal College - 1st;
 2017 International Customs Day Essay Competition - 1st;
 National Children Day Essay Competition - 1st;
 2016 Bangladesh Association for the Aged and Institute of Geriatric Medicine Essay Competition - 2nd;
 TIB Essay Competition - Honourable Mention
 2015 Essay Competition on National Mourning Day - 1st Runner Up;
 National Wash Story Writing Contest - 2nd Runner Up;
 2014 Essay Competition on National Energy Security - 3rd;
 Nazrul Institute Essay Competition - 2nd;
 Dhaka Imperial College Essay Competition - 1st;

Olympiads:

2019 1st LCGHS Inter School and College Fair '19 - Extempore Speech - 2nd; Earth Olympiad - 1st, Math - 3rd.
 2018 Holy Cross College 17th Inter College Science Festival '18 - Chemistry Olympiad - 2nd;
 2017 1st DGBHS Inter School Science Festival '17 - Chemistry Olympiad - 2nd, Extempore Speech - 1st;
 10th ISTARC Science and Technology Festival '17 - Extempore Speech - 1st, Nature Olympiad - 1st;
 2015 Junior Science Olympiads '15 (Regional) - 1st;

Academic Scholarships:

2022 Dean's Award For Academic Excellence;
 2015-2016 The Scholars Forum;
 2014-2015 Haji Shahab Uddin Memory Scholarship;
 2013-2016 The Students Welfare Foundation;
 2013, 2015 Sher-E-Bangla Memorial Scholarship;
 2013 Nur MD. Sir Memory Scholarship, BICAL Foundation;
 2011-2016 Bangladesh Cadet Academy.

LEADERSHIP EXPERIENCE

1. Campus Leader, Interactive Cares	May 2025 - Sep 2025
2. Dir. of Outreach Management, Bangladeshi Women in Computer Science and Engineering	Apr 2025 - Present
3. Assistant General Secretary	Dec 2024 - Present
Membership Development Coordinator	Jan 2024 - Dec 2024
IEEE BUET Student Branch WIE Affinity Group	
4. Logistic Secretary	Sep 2024 - Present
Information Secretary	Apr 2023 - Sep 2024
BUET Debating Club	
5. Campus Ambassador, IEEE Xtreme 18.0	May 2024 - Mar 2025
6. Campus Ambassador, Youth Opportunities	Jan 2023 - Dec 2023
7. Campus Ambassador, AppLink	Oct 2022 - Dec 2023
8. Co-ordinating Secretary, Holy Cross College English Language Club	Aug 2019 - Apr 2020

VOLUNTEER EXPERIENCE

- **Volunteer** Feb 2024 - present
Agami Education Foundation 
 - Working on "[Library Project](#)" for underprivileged, distributed books for [AAlok Shikkhaloy](#), and judged [IET 2024](#).
- **Volunteer** Oct 2020 - Dec 2023
Evolution 360 
 - Created campaign content, online in charge "[Shreyasi](#)", and joined Project [YED](#) by [MJF](#).
- **Volunteer** Dec 2022 - Jan 2024
FOLV - Forum Of Liberal Volunteers 
 - Supported "[USHNOTA](#)" project, raising funds for winter clothing.

REFERENCES

1. **Dr. A.B.M Alim Al Islam**
Professor, Department of Computer Science & Engineering
Bangladesh University of Engineering and Technology
Email: razi_bd@yahoo.com
Relationship: [Advisor, Research Supervisor]
2. **Dr. M. Sohel Rahman**
Professor, Department of Computer Science & Engineering
Bangladesh University of Engineering and Technology
Email: msrahman@cse.buet.ac.bd
Relationship: [Research Supervisor]